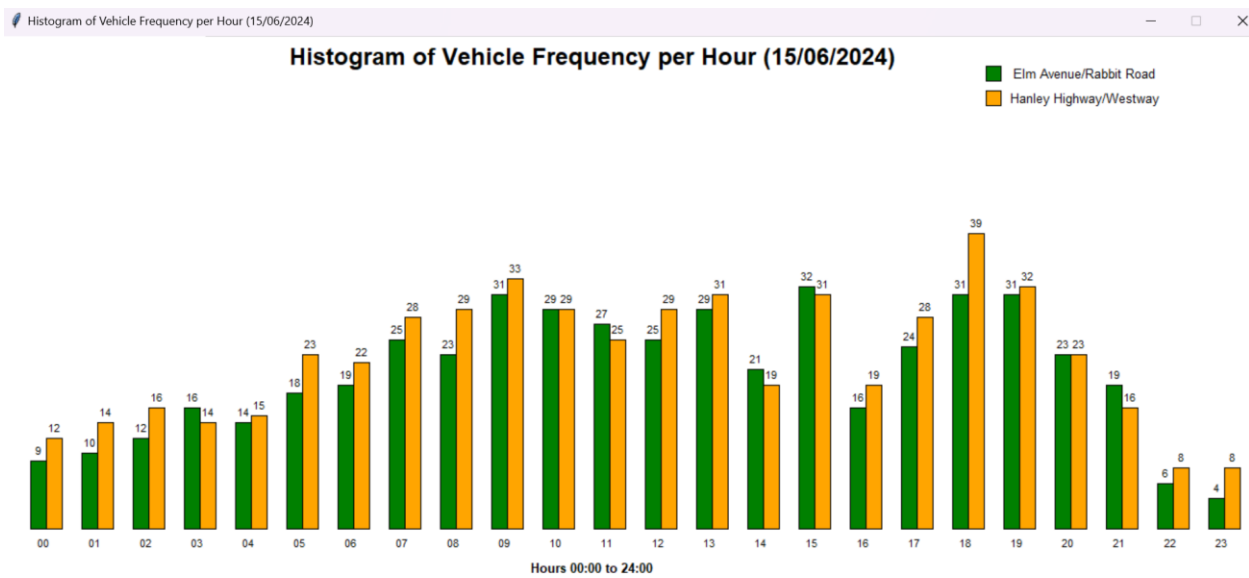


Test Case	Test Scenario	Input Data	Expected Result	Actual Result	Status(Fail/Pass)
Display histogram	Display the histogram with the traffic data according to the given date	Hourly vehicle counts of Elm Avenue / Rabbit Road and Hanley Highway/Westway	• Display the histogram in the canvas • The hour label should display below the histogram • The bars should visualize junctions with separated colours • The heights of bars should correspond to the traffic counts. • The legends should display the corresponding junctions	The all expectations can see in the canvas: Hour label Scaled bars Legends	Pass
Load the file and process the data	Verify the code correctly loads and processes the traffic data from csv file.	The valid csv file path.	• The program needs to accurately read the CSV file and convert the data into a dictionary format for traffic_data. • The data should be properly analyzed, with traffic counts for each hour and each junction stored in the dictionary.	The program should correctly load the specific data file, analyze it and store necessary data in the dictionary.	Pass

Invalid date input	Verify the program handles invalid date formats	Invalid date input like non integers and also non existing date or inputs which are not in the expected range	- The code prompts for a valid date . - If the date is invalid program should display error message with asking a valid date input	Program prompts for a valid date input and show the error message if input is invalid.	Pass
File not found	Handle the program if the given file path can't find	Non-existent file path	The program should display error message if you enter non-existing file path	The program should display error message if you enter non-existing file path	Pass
Continuation	Verify the user wants to get a new data set	Give "Y / N/y/n" as the input	If you input: - Y/y = ask another date for load the specific file - N/n = break the program - Except Y/N/y/n = display error message to only input Y/N/y/n	Prompt user for asking "do you need another data set ?" according to the given input ,expected result is happen.	Pass

Test Case 01-Display Histogram



Test Case 02 - Load the file and process the data

```
Please enter the day of the survey in the format DD: 15
Please enter the month of the survey in the format MM: 6
Please enter the year of the survey in the format YYYY: 2024
*****
Data file selected: C:/Users/ISIRI/Music/Music/course work sd/traffic_data15062024.csv
*****
The total number of vehicles recorded for this date is 1037
The total number of trucks recorded for this date is 109
The total number of electric vehicles for this date is 368
The total number of two-wheeled vehicles for this date is 401
The total number of busses leaving Elm Avenue/Rabbit Road heading North is 11
The total number of vehicles through both junctions not turning left or right is 363
The percentage of total vehicles recorded that are trucks for this date is 11%
The average number of bikes per hour for this date is 7
The total number of vehicles recorded as over the speed limit for this date is 205
The total number of vehicles recorded through Elm Avenue/Rabbit Road junction is 494
The total number of vehicles recorded through Hanley Highway/Westway junction is 543
10% of vehicles recorded through Elm Avenue/Rabbit Road are scooters.
The highest number of vehicles in an hour on Hanley Highway/Westway is 39
The most vehicles through Hanley Highway/Westway were recorded between 18:00 and 19:00
The number of hours of rain for this date is 0
```

Test Case 03 – Invalid Date Input

```
Please enter the day of the survey in the format DD: 15
Please enter the month of the survey in the format MM: 3
Please enter the year of the survey in the format YYYY: 2004
Error: The file 'C:/Users/ISIRI/Music/Music/course work sd/traffic_data15032004.csv' was not found.
Do you want to select a data file for a different date? (Y/N): |

Please enter the day of the survey in the format DD: 32
Out of range - values must be in the range 1 and 31.
Please enter the day of the survey in the format DD: |

Please enter the day of the survey in the format DD: 15
Please enter the month of the survey in the format MM: 14
Out of range - values must be in the range 1 to 12.
Please enter the month of the survey in the format MM:

Please enter the day of the survey in the format DD: 15
Please enter the month of the survey in the format MM: 6
Please enter the year of the survey in the format YYYY: 1990
Out of range - values must range from 2000 and 2024.
Please enter the year of the survey in the format YYYY: |

Please enter the day of the survey in the format DD: rhhhhy@
Integer required!
Please enter the day of the survey in the format DD: |
```

Test Case 04 - file Not Found

```
Please enter the day of the survey in the format DD: 15
Please enter the month of the survey in the format MM: 6
Please enter the year of the survey in the format YYYY: 2004
Error: The file 'C:/Users/ISIRI/Music/Music/course work sd/traffic_data15062004.csv' was not found.
Do you want to select a data file for a different date? (Y/N): |
```

Test Case 05 – Continuation

```
Please enter the day of the survey in the format DD: 15
Please enter the month of the survey in the format MM: 6
Please enter the year of the survey in the format YYYY: 2024
*****
Data file selected: C:/Users/ISIRI/Music/Music/course work sd/traffic_data15062024.csv
*****
The total number of vehicles recorded for this date is 1037
The total number of trucks recorded for this date is 109
The total number of electric vehicles for this date is 368
The total number of two-wheeled vehicles for this date is 401
The total number of busses leaving Elm Avenue/Rabbit Road heading North is 11
The total number of vehicles through both junctions not turning left or right is 363
The percentage of total vehicles recorded that are trucks for this date is 11%
The average number of bikes per hour for this date is 7
The total number of vehicles recorded as over the speed limit for this date is 205
The total number of vehicles recorded through Elm Avenue/Rabbit Road junction is 494
The total number of vehicles recorded through Hanley Highway/Westway junction is 543
10% of vehicles recorded through Elm Avenue/Rabbit Road are scooters.
The highest number of vehicles in an hour on Hanley Highway/Westway is 39
The most vehicles through Hanley Highway/Westway were recorded between 18:00 and 19:00
The number of hours of rain for this date is 0
```

Do you want to select a data file for a different date? (Y/N): |

```
Please enter the day of the survey in the format DD: 15
Please enter the month of the survey in the format MM: 6
Please enter the year of the survey in the format YYYY: 2024
*****
Data file selected: C:/Users/ISIRI/Music/Music/course work sd/traffic_data15062024.csv
*****
The total number of vehicles recorded for this date is 1037
The total number of trucks recorded for this date is 109
The total number of electric vehicles for this date is 368
The total number of two-wheeled vehicles for this date is 401
The total number of busses leaving Elm Avenue/Rabbit Road heading North is 11
The total number of vehicles through both junctions not turning left or right is 363
The percentage of total vehicles recorded that are trucks for this date is 11%
The average number of bikes per hour for this date is 7
The total number of vehicles recorded as over the speed limit for this date is 205
The total number of vehicles recorded through Elm Avenue/Rabbit Road junction is 494
The total number of vehicles recorded through Hanley Highway/Westway junction is 543
10% of vehicles recorded through Elm Avenue/Rabbit Road are scooters.
The highest number of vehicles in an hour on Hanley Highway/Westway is 39
The most vehicles through Hanley Highway/Westway were recorded between 18:00 and 19:00
The number of hours of rain for this date is 0
```

Do you want to select a data file for a different date? (Y/N): y
Please enter the day of the survey in the format DD: |

```
Please enter the day of the survey in the format DD: 15
Please enter the month of the survey in the format MM: 6
Please enter the year of the survey in the format YYYY: 2024
*****
Data file selected: C:/Users/ISIRI/Music/Music/course work sd/traffic_data15062024.csv
*****
The total number of vehicles recorded for this date is 1037
The total number of trucks recorded for this date is 109
The total number of electric vehicles for this date is 368
The total number of two-wheeled vehicles for this date is 401
The total number of busses leaving Elm Avenue/Rabbit Road heading North is 11
The total number of vehicles through both junctions not turning left or right is 363
The percentage of total vehicles recorded that are trucks for this date is 11%
The average number of bikes per hour for this date is 7
The total number of vehicles recorded as over the speed limit for this date is 205
The total number of vehicles recorded through Elm Avenue/Rabbit Road junction is 494
The total number of vehicles recorded through Hanley Highway/Westway junction is 543
10% of vehicles recorded through Elm Avenue/Rabbit Road are scooters.
The highest number of vehicles in an hour on Hanley Highway/Westway is 39
The most vehicles through Hanley Highway/Westway were recorded between 18:00 and 19:00
The number of hours of rain for this date is 0
```

Do you want to select a data file for a different date? (Y/N): 56yy
Invalid input. Please enter 'Y' to continue or 'N' to quit.

```
Please enter the day of the survey in the format DD: 15
Please enter the month of the survey in the format MM: 6
Please enter the year of the survey in the format YYYY: 2024
*****
Data file selected: C:/Users/ISIRI/Music/Music/course work sd/traffic_data15062024.csv
*****
The total number of vehicles recorded for this date is 1037
The total number of trucks recorded for this date is 109
The total number of electric vehicles for this date is 368
The total number of two-wheeled vehicles for this date is 401
The total number of busses leaving Elm Avenue/Rabbit Road heading North is 11
The total number of vehicles through both junctions not turning left or right is 363
The percentage of total vehicles recorded that are trucks for this date is 11%
The average number of bikes per hour for this date is 7
The total number of vehicles recorded as over the speed limit for this date is 205
The total number of vehicles recorded through Elm Avenue/Rabbit Road junction is 494
The total number of vehicles recorded through Hanley Highway/Westway junction is 543
10% of vehicles recorded through Elm Avenue/Rabbit Road are scooters.
The highest number of vehicles in an hour on Hanley Highway/Westway is 39
The most vehicles through Hanley Highway/Westway were recorded between 18:00 and 19:00
The number of hours of rain for this date is 0

Do you want to select a data file for a different date? (Y/N): n
Exiting the program. Thank you!
```

